

Essays on Applied Econometrics

Dissertation Defense

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Overview

- ➊ **Targeted Credit Access and Entrepreneurship**
Can targeted financial access, via specialized lender programs, improve entrepreneurial activity in underserved communities?
- ➋ ***Ce modèle vous plaît?***
How can experimental preference data on LLMs be leveraged to design cost-efficient, heterogeneity-aware routing policies?
- ➌ **Single-Shot Effects and Semiparametric DiD**
How can we identify and estimate impulse-response parameters in DiD settings with time-varying treatments and covariates?

Outline of Talk

- 1 Chapter 1: Targeted Credit Access and Entrepreneurship
- 2 Chapter 2: *Ce modèle vous plaît?*
- 3 Chapter 3: Single-Shot Effects and Semiparametric DiD
- 4 Conclusion

Entrepreneurship Gaps in the United States

- **Persistent gaps in entrepreneurship:** Business formation remains substantially lower among minority and low income groups and neighborhoods (Fairlie 2022)
- **Capital access frictions as a central mechanism:**
 - ▶ **Discrimination and banking access** Differential loan approval rates and credit rationing (Blanchflower et al. 2003; Chatterji and Seamans 2012; Li 2022)
 - ▶ **Credit histories and expectations** Weak credit worthiness, collateral, and access perceptions (Fairlie 2022; Bennett 2024)
- **Why it matters:** Financial frictions limit entry of high ability potential entrepreneurs, reduce firm scale and survival, and constrain inclusive economic growth

Motivated Policies

- Policymakers aim to ease capital access constraints to unlock local employment and neighborhood development by enabling entry and growth of constrained entrepreneurs.
- **Common tools:**
 - ▶ **Disclosure regulations** - e.g., Community Reinvestment Act
 - ▶ **Credit guarantees and subsidies** - e.g., Small Business Administration loan subsidies (504 and 7a programs)
- **Limitations:** Disclosures influence behavior only indirectly, guarantees and tax incentives may not be tightly targeted to the most constrained or highest return entrepreneurs
 - ▶ Hackney et al. 2023; Ding and Lee 2018

Targeted Place-Based Lending via CDFIs

- **A targeted, place-based approach:** Expand credit access via mission-driven lenders that specialize in constrained borrowers and disadvantaged neighborhoods.
- **Community Development Financial Institutions (CDFIs):**
 - ▶ Increases credit supply by certifying and subsidizing community-oriented lenders
 - ▶ Intermediaries include nonprofit banks, credit unions, and loan funds operating in designated markets
- **What CDFIs provide:**
 - ▶ **Targeted lending:** majority of financing directed to defined populations and places
 - ▶ **Diverse intermediary services:** consumer, housing, small business, and venture finance; advisory and technical assistance

This Paper

- **Objective:** Evaluate how local access to CDFIs affects the rate, composition, and early outcomes of new startups
- **Motivation:** Nearly three decades in, causal evidence on CDFI program effects remains scarce (Harger et al. 2015; Johnson et al. 2025; Gong et al. 2023; Swack et al. 2015)
- **Empirical Strategy:** Employ a difference-in-differences design using ZIP-level entrepreneurship outcomes, with treatment defined as the observed entry of CDFIs from disclosure reports
- **Contributions:**

Ch. 1: Targeted Credit Access and Entrepreneurship

Background

CDFI Program

- Established under the 1994 Riegle Community Development and Regulatory Improvement Act and overseen by the U.S. Treasury
 - ▶ First certifications and awards in 1996 with 69 lenders
- Participating lenders receive competitive grants from the CDFI Fund to support lending and technical assistance
- Certified loan funds often leverage CRA motivated partnerships with banks for supplemental capital and investment
- **Program size today:**
 - ▶ More than 1,400 certified CDFIs nationwide across all 50 states
 - ▶ Estimated \$500 billion in managed loan portfolios, as of 2023 (Anderson 2024)

Target Investment Markets

- To maintain certification, lenders must direct at least 60% of financing activity toward a designated investment market
- At certification, CDFIs designate one or more target markets and must annually demonstrate sufficient activity within them
- Target markets can be:
 - ▶ **Geographic (Investment Areas):** Census tracts that meet at least one criterion poverty rate $\geq 20\%$, median income $\leq 80\%$ of the U.S. median, or unemployment $\geq 1.5\times$ the U.S. rate
 - ▶ **Population based (Targeted Populations):** Defined borrower groups and purposes, such as Native communities, minority borrowers, or other underserved populations

Ch. 1: Targeted Credit Access and Entrepreneurship

Data

Data Sources

Balanced panel of $\approx 32,000$ ZIP codes from 1988 to 2016

① CDFI Fund Transaction Level Reports (TLR)

- ▶ Loan-level data on CDFI activity after CDFI Fund grant receipt

② Startup Cartography Project

- ▶ ZIP code measures of entrepreneurial activity based on state business filings and prediction models
- ▶ Includes corporations and partnerships, excluding non-profits, non-employer, and sole-proprietor startups

③ Demographic and other control variables

- ▶ Decennial Census, National Neighborhood Data Archive, ZIP Code Business Patterns

Outcomes of Interest

- **Startup formation**

- ▶ (Flow) Startup formation rate: new firms per 1,000 residents
- ▶ (Stock) Number of Establishments per 1,000 residents

- **Startup quality**

- ▶ Average predicted startup quality score from the Startup Cartography Project models

- **Growth outcomes**

- ▶ Counts of high growth events such as IPOs and acquisitions within 6 years of founding among local startups

- **Labor market outcomes**

- ▶ Employment-to-Population ratios and log wages

Ch. 1: Targeted Credit Access and Entrepreneurship

Empirical Strategy

Local CDFI Access and Startup Activity

- Study how local startup formation responds when neighborhoods gain access to at least one CDFI lender
- Use a difference in differences design with staggered timing of CDFI entry across counties
 - ▶ Infer CDFI entry from the first appearance of TLR lending
 - ▶ Assign treatment cohorts at the county level
 - ▶ Study outcomes at ZIP level, study demographic heterogeneity

Ideal Reduced Form Model

Two-Way Fixed Effects:

$$Y_{ict} = \alpha_{ic} + \delta_t + \beta \mathbb{I}[\text{ZIP } i \text{ in a CDFI's active TIM}] + \varepsilon_{ict}$$

- Y_{ict} : entrepreneurship outcome for ZIP i in county c at time t
- Treatment indicator is whether ZIP i is (at least partly) included in an existing CDFI's active investment market
- **Main Identification Challenges:**
 - 1 Do not observe individual CDFI's TIM definitions nor their exact certification/entry date
 - 2 CDFI entry is non-random and may be related to other factors that affect entrepreneurial activity

Feasible Reduced Form Model

Alternative Specification:

$$Y_{ict} = \alpha_{ic} + \delta_t + \tilde{\beta} \mathbb{I}[\text{Observed Lending in County } c \text{ at some } t' \leq t] + \tilde{\epsilon}_{ict}$$

- Define treatment as the observed ramp-up of CDFI activity in a local credit market following subsidy receipt from the CDFI Fund
 - ▶ Local credit market proxied for by county-level definition.
→ Finer definitions may mis-classify treatment
- TLR disclosures reveal all award-associated lending
 - ▶ **Operating Assumption:** first recorded areas of lending reveal lenders' approximate investment market/geographic scope
 - ▶ Over 90% of CDFIs receive one CDFI-Fund-backed award

Addressing Non-Random Entry

- Identification of $\tilde{\beta}$ may still be affected by other local factors impacting both entrepreneurship trends and CDFI entry
 - ▶ ZIPs in counties where CDFIs never enter are likely poor comparisons due to demographic differences
 - ▶ Instead, eventually-treated ZIPs come from the same policy-eligible pool, with similar institutional motivations for CDFIs to define them as a TIM
 - ▶ Because we define treatment at a coarse level (county) rather than unobserved target markets for each lender, it is important to compare neighborhoods with similar characteristics

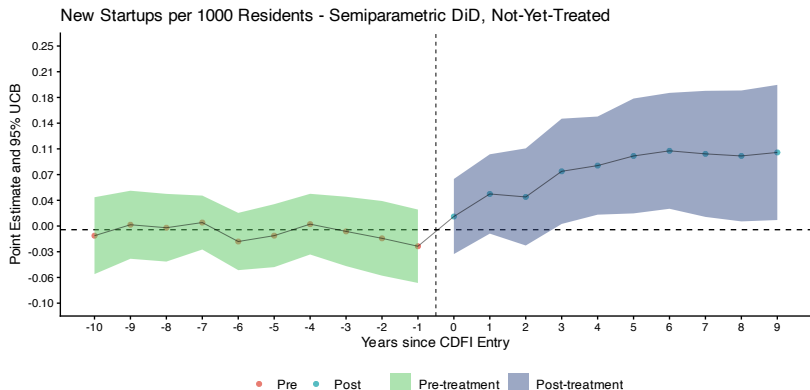
Estimation

- Estimate each $ATT(g, t)$ and aggregate to event-study effects, following Callaway and Sant'Anna (2021)
- Employ doubly-robust $ATT(g, t)$ estimators
 - ▶ Large covariate set $\rightarrow$ apply the Double Machine Learning (Chernozhukov et al. 2018) to estimate first-stage outcome regression and propensity score nuisance parameters
 - ▶ Use cross-fit random forests – better validation set performance than LASSO or LightGBM
 - ▶ Cluster at state level, report uniform confidence bands

Ch. 1: Targeted Credit Access and Entrepreneurship

Results

CDFI Entry and Startup Formation



Notes: Standard errors clustered by state. First stage estimates obtain using cross-fit random forests.

Tabulated Results

Outcome	10-Year ATT (SE)
New Startups per 1,000 Residents	0.11* (0.03)
Establishments per 1,000 Residents	0.17* (0.04)
Avg. Predicted Quality of New Startups	0.00 (0.02)
Number of Growth Events (IPO/Acquisition)	0.00 (0.01)
Employment-to-Population Ratio ($\times 100$)	0.47* (0.11)
Log Average Wages	0.00 (0.00)

Notes: * $p < 0.05$. Standard errors clustered by state.

Demographic Heterogeneity in Startup Formation

Subgroup	10-yr ATT (SE)
High Minority (bottom quartile % White)	0.12* (0.04)
Low Minority (top quartile % White)	0.06 (0.05)

Notes: Outcome is new startups per 1,000 residents. * $p < 0.05$.
Standard errors clustered by state.

Ch. 1: Targeted Credit Access and Entrepreneurship

Discussion

Discussion

- **Targeted credit works on the extensive margin.** CDFI entry brings in more founders without lowering predicted quality or realized growth outcomes.
- **The new activity is real and durable.** Establishment counts and local employment rise and persist, indicating net new business formation rather than reshuffling across places or firms.
- **Effects are largest where credit constraints bind.** Gains concentrate in high-minority, lower-income areas, consistent with the program's targeting and with models of credit-constrained entry (Evans and Jovanovic 1989).

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Large Language Models and Task Deployment

- **Context.** Organizations deploy many LLMs across heterogeneous tasks. Models differ greatly in task-specific quality and inference costs.
- **The Problem.** Which model should serve a given task?
- **Why It Matters.** Inference at scale is cost-intensive. Always routing to the best-benchmarked model can be inefficient.

This Paper

- **Objective:** Estimate *heterogeneous* LLM quality measures from experimental preference data and use it to design cost-aware routing policies
- **Approach:** A heterogeneity-enriched Bradley–Terry model whose parameters vary with prompt embeddings, estimated by structural deep learning (Farrell, Liang, and Misra 2021)
- **Routing Policy Design:** Embed model-specific quality and cost prediction functions into a planner's routing problem; learn the policy via Athey and Wager (2021), with an orthogonal score
- **Empirical Application:** 48,760 pairwise evaluations across 50 LLMs from the French government's Compar:IA platform, with per-query energy/token costs

Model Framework

Classical Bradley–Terry Model

In practice. Platforms like Chatbot Arena (Chiang et al. 2024) crowdsource *randomized pairwise* comparisons of LLM responses and rank models from the human votes via Bradley–Terry.

Standard formulation (Bradley and Terry 1952; Luce 1959). Each model m has a scalar quality parameter $\theta_m \in \mathbb{R}$. The win probability for model i over j is:

$$\Pr(Y_{rk}(i,j) = 1) = \frac{\exp(\theta_i)}{\exp(\theta_i) + \exp(\theta_j)}$$

Properties. Transitivity across alternatives. Only pairwise quality differences $\theta_i - \theta_j$ are identified $\Rightarrow$ normalize $\theta_1 = 0$.

Limitation. Preferences are **homogeneous**: relative quality does not vary across prompts, tasks, or contexts. A model that is best on average may be mediocre for specific prompts.

Heterogeneity-Enriched Bradley–Terry

Enrichment: replace the constant θ_m with a flexible function of the conversation embedding X_k (Li and Li 2025; Zhang et al. 2025).

$$\Pr(Y_{rk}(i,j) = 1 \mid X_k) = \frac{\exp\{\theta_i(X_k)\}}{\exp\{\theta_i(X_k)\} + \exp\{\theta_j(X_k)\}}$$

The functions $\{\theta_m(\cdot)\}_{m \in \mathcal{M}}$ are defined as loss minimizers

$$\{\theta_m(\cdot)\}_{m \in \mathcal{M}} = \arg \min_{\{f_m\}} \mathbb{E} \left[\ell(Y_{rk}, f_{I_{rk}}(X_k), f_{J_{rk}}(X_k)) \right]$$

where ℓ is the Bradley–Terry logistic log-likelihood.

Each embedding X_k induces *different* preference parameters and potentially *different* model rankings.

Identifiable Parameters

Average Preference Parameter:

$$\bar{\theta}_m = \mathbb{E}[\theta_m(X_k)]$$

Summarizes the average quality of model m relative to the baseline, integrating over prompt contexts.

(Average) Dominance Probability:

$$Q_m(X_k) = \frac{\exp\{\theta_m(X_k)\}}{\sum_{j=1}^M \exp\{\theta_j(X_k)\}}, \quad \bar{Q}_m = \mathbb{E}[Q_m(X_k)]$$

i.e., the probability that m is most preferred among M alternatives

Estimation and Routing Policies

Estimation: Structural Deep Learning (Farrell, Liang, and Misra 2021)

- Estimate $\{\theta_m(\cdot)\}_{m=2}^M$ with a **multi-headed neural network**:
 - ▶ *Shared trunk*: maps $X_k \in \mathbb{R}^d$ through a shared network
 - ▶ *Model-specific heads*: each head m outputs $\hat{\theta}_m(X_k) \in \mathbb{R}$
- Construct a Neyman-orthogonal score $\psi(W_{rk})$ for a given smooth functional $\mu = \mathbb{E}[G(X_k, \theta(X_k))]$ of the preference parameters (and other regression objects):

$$\psi(W_{rk}) = G(X_k, \theta(X_k)) - G_{\theta}(X_k, \theta(X_k)) \underbrace{\mathbb{E}[\ell_{\theta\theta}(Y_{rk}, \theta(X_k)) \mid X_k]^{-1}}_{\text{inverse conditional Hessian}} \ell_{\theta}(Y_{rk}, \theta(X_k))$$

- Neyman orthogonality facilitates valid statistical inference and policy learning (Athey and Wager 2021)

A Cost-Aware Routing Problem

Setup. For each conversation k with embedding X_k , the planner chooses a model to deploy.

Inference costs: $c_m(X_k) = \mathbb{E}[K_m \mid X_k]$, where K_m is the energy/tokens/money consumed by model m

Routing policy: $\pi : \mathcal{X} \rightarrow \mathcal{M}$. Maps from prompt context to model.

Planner's objective: minimize expected inference cost conditional on selecting the user-preferred model (i.e., re-weighted cost minimization):

$$\min_{\pi \in \Pi} \mathbb{E} \left[c_{\pi(X_k)}(X_k) \mid M^* = \pi(X_k) \right] = \min_{\pi \in \Pi} \mathbb{E} \left[\frac{Q_{\pi(X_k)}(X_k)}{\mathbb{E}[Q_{\pi(X_k)}(X_k)]} c_{\pi(X_k)}(X_k) \right]$$

Intuition. Route to cheaper models when dominance probability differences are small; use expensive models only when they are strongly preferred for a given prompt.

Policy Learning

Augmented loss. To construct orthogonal scores for the ratio objective, jointly estimate preferences *and* inference costs:

$$\ell^*(W_{rk}, \boldsymbol{\varphi}) = \ell_{\text{BT}}(W_{rk}, \boldsymbol{\theta}) + \sum_{m=1}^M \frac{1}{2} (K_m - c_m(X_k))^2$$

where $\boldsymbol{\varphi} = (\boldsymbol{\theta}, c)$ collects all nuisance functions and $c_m(X_k) = \mathbb{E}[K_m \mid X_k]$.

Orthogonal influence function. For each model m , construct an orthogonal score $\boldsymbol{\psi}_m^*(W_{rk})$ for $\mathbb{E}[c_m(X_k) \mid M^* = m]$ using results from Farrell, Liang, and Misra (2021).

Policy optimization (Athey and Wager 2021):

$$\max_{\pi \in \Pi} \mathbb{E} \left[\sum_{m=1}^M -\boldsymbol{\psi}_m^*(W_{rk}) \mathbf{1}\{\pi(X_k) = m\} \right]$$

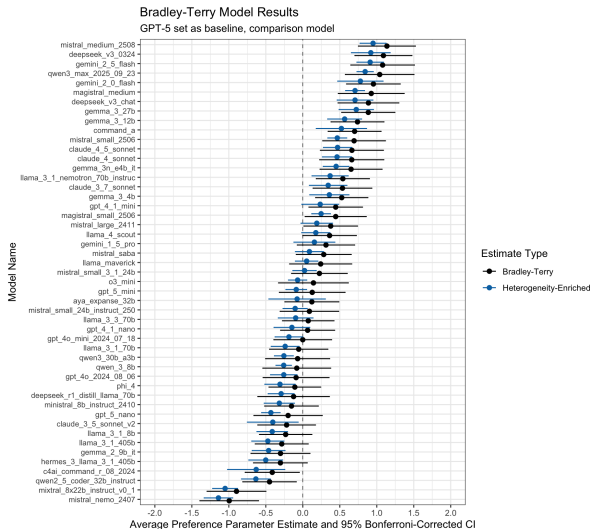
Policy class: neural network classifier with max-of-softmax output.

Empirical Application

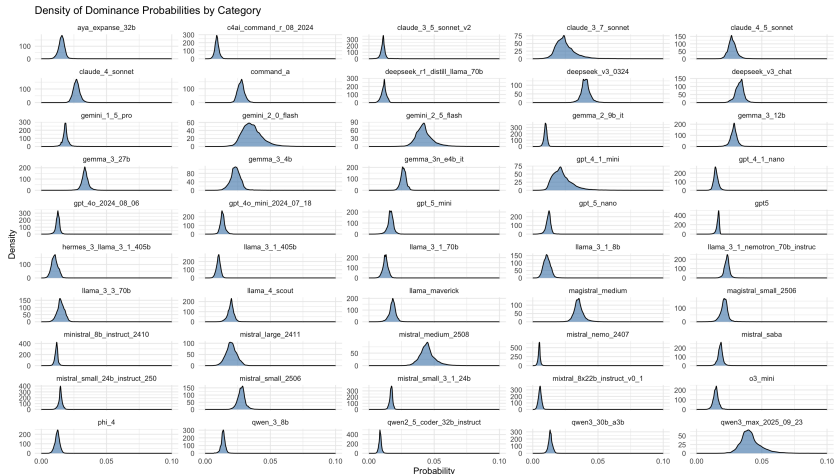
Data: The Compar:IA Platform

- **Setting.** The French government's Compar:IA platform: large-scale, randomized pairwise human evaluation of 50 LLMs, paired with model-level inference-energy costs.
- **Source.** Compar:IA, a large-scale pairwise-evaluation platform run by the French Ministère de la Culture: randomized assignment with blind outputs and post-vote reveal.
- **Sample.** 48,760 pairwise experiments (32,744 decisive); 50 most-evaluated models from 10 developers (Jan–Nov 2025).
- **Context.** Each prompt embedded into a 384-dim vector (paraphrase-multilingual-MiniLM-L12-v2), plus topic and token-length covariates.
- **Inference costs.** Per-query energy (kWh) from EcoLogits (Rincé and Banse 2025) estimates—a function of output token length and model complexity.

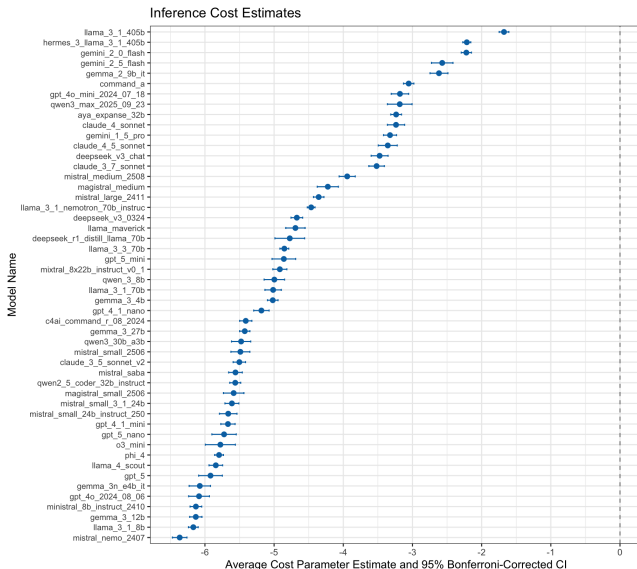
Average Bradley-Terry Estimates: Unconditional vs. Conditional



Distribution of Preference Heterogeneity by Model



Average Inference-Cost Predictions by Model



Results: Cost-Aware Routing Policy

We compare the learned cost-aware policy π^{cost} to the preference-first benchmark $\pi^{\text{pref}}(X) = \arg \max_m \hat{Q}_m(X)$. The target parameters are the policy contrasts in expected log inference cost and expected dominance:

$$\mu_{\text{Cost}} = \mathbb{E}[\log c_{\pi^{\text{cost}}}] - \mathbb{E}[\log c_{\pi^{\text{pref}}}], \quad \mu_{\text{Pref}} = \mathbb{E}[Q_{\pi^{\text{cost}}}] - \mathbb{E}[Q_{\pi^{\text{pref}}}].$$

Policy contrast (cost-aware – preference-first)	Estimate	SE
μ_{Cost} (expected log inference cost)	-0.816*	(0.041)
μ_{Pref} (expected dominance probability)	+0.001	(0.002)

- $\mu_{\text{Cost}} = -0.82$ is a $\approx 80\%$ reduction in expected inference energy, while μ_{Pref} is statistically indistinguishable from zero.

$N = 32,744$. Judge-clustered standard errors. * $p < 0.05$.

Discussion

Discussion

- **A structural bridge for AI evaluation.** The first application of structural deep learning (Farrell, Liang, and Misra 2021) to preference data takes Bradley–Terry as primitive and recovers context-specific quality with valid orthogonal inference.
- **From rankings to routing.** Estimated dominance probabilities and cost predictions feed an Athey and Wager (2021) policy-learning problem with a tailored orthogonal score, turning leaderboard evaluation into deployable, statistically-backed rules.
- **Gains at negligible downside.** Cost-aware routing cuts expected inference energy by about 80 percent, with a quality change indistinguishable from zero.

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Motivation

- Some empirical settings involve time-varying shocks, where overall exposure to an intervention increases over time
 - ▶ e.g., investment/tax incentives, minimum wages, pollution
- It can be challenging to summarize **dynamic** treatment responses due to multiple (continuous) dose exposures
- To inform policy designs, it is often useful to understand causal responses to a single treatment increment in addition to the overall program exposure.
- Conventional strategies in the DiD literature summarize aggregate responses to the entire path of treatment levels or focus on static, one-period-ahead effects
 - ▶ de Chaisemartin de Chaisemartin et al. 2022, 2024, 2025

This Paper

- ① Develops identification results for dynamic, single-shot treatment effects using difference-in-differences under conditional-on-observables parallel trend assumptions
 - ▶ Impulse response to the first non-zero dose over time
- ② Constructs two-step estimators that may incorporate machine learning methods for high-dimensional covariate settings
- ③ Applies the estimators to an empirical application on the labor market effects of the Small Business Administration's 504 loan guarantee program

Framework

Setup

We present results for the 3-period ($T = 3$) case with a single cohort, but generalize to more periods and staggered adoption in the paper.

- Let L_{it} denote unit i 's treatment dose in period t
- All units are untreated in the first period ($L_{i1} = 0$)
- A group of treated units receives a first dose in period 2, $L_{i2} \in \mathbb{R}_+$, and a potential second dose in period 3, $L_{i3} \in \mathbb{R}_{\geq 0}$.
- There exists a Never Treated group: $L_{i1} = L_{i2} = L_{i3} = 0$

Potential Outcomes for Cumulative Exposure

- Let $D_{it} = \sum_{l=1}^t L_{il}$ denote the cumulative exposure to the treatment intervention as of period t
- Let $\vec{D}_i = (D_{i1}, D_{i2}, D_{i3}) \in \mathbb{R}_{\geq 0}^3$ be the assigned vector of exposures
 - ▶ Never Treated units associated with the zero vector, $\vec{0}$
 - ▶ Let $G_i = \mathbb{I}(\vec{D}_i \neq \vec{0})$ denote treatment group assignment
- Potential outcomes defined over exposure paths:

$$Y_{it}(\vec{D}_i) = Y_{it}(D_{i1}, D_{i2}, D_{i3})$$

Conventional Estimand: Actual vs Status Quo

- ① Actual-vs-Status-Quo ATT , 2-periods from initial treatment:

$$\begin{aligned}\theta^{ATT}(2) &= \mathbb{E} \left[Y_{i3}(0, D_{i2}, D_{i3}) - Y_{i3}(\vec{0}) \mid G_i = 1 \right] \\ &= \mathbb{E} \left[Y_{i3} - Y_{i3}(\vec{0}) \mid G_i = 1 \right]\end{aligned}$$

- ▶ Aggregates causal responses to all prior changes in treatment
- ▶ Target parameter with conventional strategies (de Chaisemartin and d'Haultfoeuille 2024)

Our Parameter of Interest: Single-Shot

- ① Single-Shot *ATT*, 2-periods from initial treatment:

$$\theta^{SS}(2) = \mathbb{E} \left[Y_{i3}(0, D_{i2}, D_{i2}) - Y_{i3}(\vec{0}) \mid G_i = 1 \right]$$

- ▶ Our parameter of interest, but not as easily identified as θ^{ATT}
- ▶ Treated units may receive more than one treatment across the panel, with each dose varying in size, so we don't observe the counterfactual

Identification

Identification Strategy

- We use a **(quasi-)stayer** identification strategy:
 - ▶ We assume no treated unit remains at their first exposure, but a non-zero proportion of units remain within a small bandwidth of what would be their single-shot treatment path
- Extension of Stayer/Quasi-Stayer frameworks of:
 - ▶ de Chaisemartin & d'Haultfoeuille (2024a): continuous-treatment DiD with stayers
 - ▶ de Chaisemartin & d'Haultfoeuille (2024b): intertemporal treatment effects
- All results require an untreated comparison group to proxy for the unexposed counterfactuals of the treated group

Identification Assumptions

Define X_i as a (potentially high-dimensional) vector of time-invariant baseline characteristics for each unit.

Assumption 1 (Overlap). With f the conditional density of period-2 treatments,

$$\mathbb{P}(\vec{D}_i = \vec{0} \mid X_i) < 1, \quad f(D_{i2} = d_2 \mid G_i = 1, X_i) > 0 \quad \forall d_2 \in \mathbb{R}_+$$

Ensures the conditional means used to identify θ^{SS} are well-defined.

Assumption 2 (Conditional Parallel Trends).

$$\forall \vec{d} \neq \vec{0}: \mathbb{E}[Y_{i3}(\vec{0}) - Y_{i1}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \vec{d}] = \mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 0, X_i]$$

Never-treated conditional trends serve as the counterfactual for similar-looking treated units.

Identification Assumptions (Continued)

Assumption 3 (“Strong” Conditional Parallel Trends). For any $\vec{d}, \underline{d} \in \mathbb{R}_{\geq 0}^3$ with $\vec{d} \neq \underline{d}$,

$$\mathbb{E}[Y_{i3}(\vec{d}) - Y_{i3}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \vec{d}] = \mathbb{E}[Y_{i3}(\vec{d}) - Y_{i3}(\vec{0}) \mid G_i = 1, X_i, \vec{D}_i = \underline{d}]$$

Conditional on observables, no selection on treatment effects into particular exposure paths.

Assumption 4 (Existence of Conditional Quasi-Stayers).

$$\mathbb{P}(|D_{i3} - D_{i2}| < \eta \mid G_i = 1, X_i, D_{i2}) > 0, \quad \forall \eta > 0$$

Some treated units remain within a small bandwidth of their single-dose path.

Identification Results

Under Assumptions 1-3 and 4, $\theta^{SS}(2)$ is identified by

$$\theta^{SS}(2) = \mathbb{E} \left[\lim_{\eta \rightarrow 0} \mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 1, X_i, D_{i2}, |D_{i3} - D_{i2}| < \eta] \mid G_i = 1 \right] \\ - \mathbb{E}[\mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 0, X_i] \mid G_i = 1] \quad (1)$$

Estimation

Parametric Estimation Strategy with Quasi-Stayers

- In this draft, we propose parametric, $\sqrt{n}$ -consistent estimators based on conditional mean specifications:

Assumption 5: Parametric Model

$$\mathbb{E}[Y_{i3} - Y_{i1} \mid G_i = 1, \vec{D}_i, X_i] = g(\vec{D}_i, X_i; \omega)$$

- For instance, $g(\vec{D}_i, X_i; \omega) = \alpha + \vec{D}_i^T \beta + X_i^T \lambda$

More Flexible Parameterization

- Varying Coefficient Model:

$$\tilde{g}(\vec{D}_i; \omega(G_i, X_i)) = q_L(\vec{D}_i)^T \beta(G_i, X_i)$$

- Apply ML techniques for estimating these first-stage models (e.g., neural nets, LASSO)
- Use debiasing techniques from the double machine learning literature (Chernozhukov et al. 2021; Chernozhukov et al. 2018)

Automatic Debiasing for ML Estimation

- Let $W_i = (Y_{i3}, Y_{i1}, G_i, \vec{D}_i, X_i)$. If $\tilde{g}$ is linear, we may write $\theta^{SS}(2)$ as a linear functional of first-stage regression functions:

$$\theta^{SS}(2) = \mathbb{E}[\tilde{g}((0, D_{i2}, D_{i2}); \omega(1, X_i)) - \tilde{g}(\vec{0}; \omega(0, X_i)) \mid G_i = 1] = \mathbb{E}[H(W_i, \tilde{g})]$$

- Need to account for first-stage estimation biases from ML — Automatic Debiasing (“AutoDML”) (Chernozhukov et al. 2021)
- Estimate α_2^{RR} and $\tilde{g}$, construct the orthogonal influence function:

$$\psi_2 = H(W_i, \tilde{g}) + \alpha_2^{RR}(W_i) \times \left(Y_{i3} - Y_{i1} - \tilde{g}(\vec{D}_i; \omega(G_i, X_i)) \right)$$

- No need to derive the influence function, other than knowing $H(W_i, \tilde{g})$. Use a simple loss to obtain α_2^{RR} :

$$\min_{a \in \mathcal{A}} \sum_{i=1}^n -2H(W_i, a) + (a(W_i))^2$$

Empirical Application

Empirical Application: SBA 504 Program

- Analyze the regional labor market effects of SBA 504 lending to large-scale local capital investment projects
 - ▶ **Previous Work:** county-level, or firm-level analysis (Brown et al. 2017; Craig, Jackson, and Thomson 2009)
 - ▶ **Analysis:** Study ZIP code employment trends after post-recession (2010-2019) 504 investment
 - ▶ **Challenge:** Multiple within-region 504 investments over time (3 projects of varying size—each \$1.6M, on average)
 - ▶ Estimate the single-shot employment impacts, and compare to conventional DiD strategies; specify treated outcome trends as varying coefficients in cumulative dose increments:

$$\mathbb{E}[Y_{i,g+\ell} - Y_{i,g-1} \mid G_i = g, X_i, \vec{D}_i] = \beta_0(X_i) + \sum_{k=0}^{\ell} \beta_k(X_i)(D_{i,g+k} - D_{i,g+k-1})$$

Results

Main Single-Shot Results

Table: Log Employment, 5-Year Estimates

Model	Estimate	Conf. Int.	% Overall
AVSQ	0.0266	[0.0172, 0.0359]	
Single-Shot	0.0114	[0.0035, 0.0193]	43%

Local Multiplier Results (Jobs per Jobs Created)

Normalized single-shot (jobs per 504-backed job):

$$\theta^{SS, \text{Norm}}(g, \ell) = \frac{\theta^{SS}(g, \ell)}{\mathbb{E}[D_{ig} \mid G_i = g]}$$

Normalized AVSQ (de Chaisemartin et al. 2024):

$$\theta^{AVSQ, \text{Norm}}(g, \ell) = \frac{\theta^{AVSQ}(g, \ell)}{\theta^D(g, \ell)}, \quad \theta^D(g, \ell) = \mathbb{E} \left[\frac{1}{\ell + 1} \sum_{s=g}^{g+\ell} D_{is} \mid G_i = g \right]$$

Table: Local Multipliers on Manufacturing Employment, 5-Year Estimates

Estimator	Estimate	95% Conf. Int.
de Chaisemartin & d'Haultfoeuille 2024	0.98	[0.73, 1.23]
Single-shot	2.37	[1.90, 2.83]

Discussion

Discussion

- **Methodology.** The single-shot estimand isolates the dynamic effect of a unit's first dose, which conventional DiD blurs by cumulating the entire exposure path.
- **Identification and Estimation.** A quasi-stayer strategy paired with AutoDML debiasing, gives doubly-robust root- n inference under high-dimensional covariates.
- **Application.** One average 504 project raises local employment about 1.1 percent over five years, roughly 43 percent of the AVSQ effect. Manufacturing employment multipliers understated by standard approaches.

Outline of Talk

- 1 Chapter 1: Targeted Credit Access and Entrepreneurship
- 2 Chapter 2: *Ce modèle vous plaît?*
- 3 Chapter 3: Single-Shot Effects and Semiparametric DiD
- 4 Conclusion

What's Next?

- Santa Barbara → San Diego ✓
- Work as a Senior Data Scientist at Intuit ✓
 - ▶ Analyzing experiments (over 50!) relating to product strategy, promotional offers, and pricing
 - ▶ Designing experimental plans for our next year offer strategies
 - ▶ Extending existing price and targeted discount models using structural models and ML
 - ▶ Leveraging AI in almost every task I do as an economist

Thank You

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CDFI Transaction Level Reports (TLR)

- Primary transaction-level source on CDFI lending since 1996
- **What is reported**
 - ▶ Every loan or investment linked to a CDFI Fund award over a 3 year reporting window
 - ▶ Fields include anonymized awardee ID, year, amount, product type, sector, and borrower address (geocoded to ZIP)

Inferring CDFI Entry from TLR

- **Use of TLR**

- ▶ Treat the first year in which a county records any CDFI TLR lending as the onset of local CDFI access
- ▶ More than one CDFI may be active in a county across the panel
- ▶ Non-awarded CDFIs may be active but not visible in the TLR

- **Interpretation of “entry”**

- ▶ Entry captures the first observable presence of CDFI Fund supported lending in an area
- ▶ Coarse measure of CDFI access and may understate earlier, potentially smaller scale, activity prior to award receipt.

Motivating County-Level Assignment

Average, Square-Approximate County Width: 40 miles

